

Assignment 1

a) In an experiment examining the effects Tai Chi on arthritis pain, Callahan (2010) selected a large sample of individuals with doctor-diagnosed arthritis. Half of the participants immediately began a Tai Chi course and the other half (the control group) waited 8 weeks before beginning. At the end of 8 weeks, the individuals who had experienced Tai Chi had less arthritis pain than those who had not participated in the course.

- i. Identify the independent variable for this study.
- ii. What scale of measurement is used for the independent variable?
- iii. Identify the dependent variable for this study.
- iv. What scale of measurement is used for the dependent variable?

(b)

Two scores, X and Y, are recorded for $n = 4$ subjects. For these scores, find the value of each expression.

subject	X	Y
A	3	4
B	0	7
C	-1	5
D	2	2

- i. $\Sigma(X-1)$
- ii. $\Sigma Y + 2$
- iii. $\Sigma(X+1)(Y-2)$

(c) An instructor has obtained the $N = 25$ exam scores shown here.

82, 75, 88, 93, 53, 84, 87, 58, 72, 94, 69, 84, 61, 91, 64, 87, 84, 70, 76, 89, 75, 80, 73, 78, 60

Construct a grouped frequency table

(d) If the following scores are ranked, what rank is assigned to an individual with a score of $X = 7$?

Scores: 1, 2, 2, 5, 6, 6, 6, 7, 10, 12

e) The following data of x =rainfall volume and y =runoff volume for a particular location. The accompanying values were read from the plot.

x	5	12	14	17	23
y	4	10	13	15	25

i) Draw a scatterplot of the data, does it support the use of the simple linear regression model?

ii) Calculate point estimates of the slope and intercept of the population regression line.

iii) Calculate a point estimate of the true average runoff volume when rainfall volume is 50.

f) Given the following data summaries

$n=18$, $\sum x_i = 1950$, $\sum y_i = 47.92$, $\sum x_i^2 = 251970$, $\sum y_i^2 = 130.6074$, $\sum x_i y_i = 5530.92$

Calculate PPMCC

g) The flow rate y (m^3/min) in a device used for air-quality measurement depends on the pressure drop x (inches of water) across the device's filter. Suppose that for x values between 5 and 20, the two variables are related according to the simple linear regression model with a true regression line

$$y = -0.12 + 0.095x.$$

i) What is the expected change in flow rate associated with a 1-inch increase in pressure drop?

- ii) What change in flow rate can be expected when pressure drop decreases by 5 inches?
- iii) What is the expected flow rate for a pressure drop of 10 inches?